

LAB 1

TECH 60 - Sec 14

F26

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Objective:

The objective for this lab was to create a circuit which included a 330 ohm resistor, and a Red LED. The circuits were created both online using Multisim, and physically using class equipment. The results from this experiment showed that Kirchhoff's Law is valid, and that the voltage across the resistor, and LED, approximately add up the source voltage.

Equipment/Components:

- Multisim
- Red LED
- 330 Ohm resistor
- Button
- Jumper wires
- 2 Banana to mini grabber cables
- Digital Multimeter
- DC Power Supply

Data:

Part 2. Hand calculation of the circuit parameters

$V_s = 9V$, $R = 330 \Omega$, Red LED

$V_{LED} = 1.85 V$

Kirchhoff's Law

$V_s = V_{LED} + V_R$

$V_R = V_s - V_{LED}$

$V_R = 9 - 1.85$

$V_R = 7.15V$

Part 3. Circuit Simulation

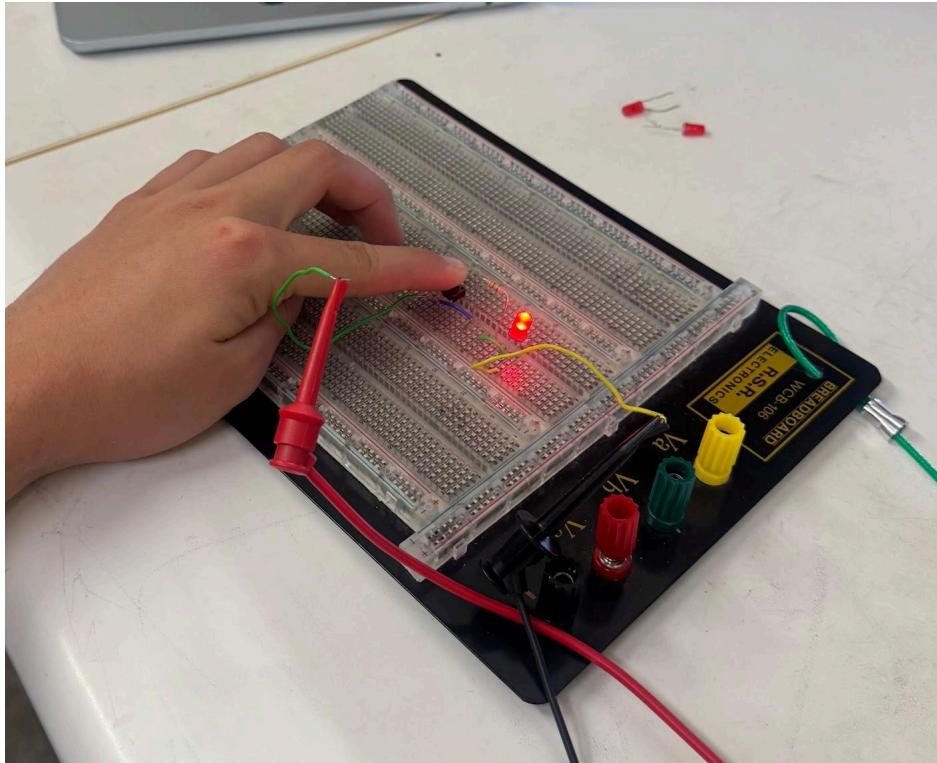
The image displays two screenshots of a Multisim circuit simulation environment, showing a circuit with a 9V battery (V1), a switch (S1), a 330Ω resistor (R1), and an LED (LED1).

Top Screenshot: The circuit is shown in the Design Toolbox. The components are connected in a series loop: V1 (9V) is connected to S1 (Key = Space), which is connected to R1 (330Ω), which is connected to LED1. The circuit is powered on, and the LED is illuminated.

Bottom Screenshot: The same circuit is shown, but with a digital multimeter (XMM1) connected in parallel with the LED. The multimeter displays a voltage of 1.83 V. The circuit is powered on, and the LED is illuminated.

The simulation results are displayed in the bottom status bar, showing the time as 32.816 s and the temperature as 74°F.

Part 4.



Voltage across resistor



Voltage across LED



	V_s	V_R	V_{LED}	Confirm $V_s = V_R + V_{LED}$
Hand Calculation	9V	7.15V	1.85V	$9V = 9V$
Multisim	9V	7.17V	1.83V	$9V = 9V$
Breadboard	8.46V	6.37V	2.086V	$9V \neq 8.46V$
%Error (PE)	6%	11.16%	13.98%	N/A

Part 5.

1: What do you observe about the relationship between the source voltage, V_s , V_R , and V_{LED} . Write an expression or few words to explain the relationship.

The voltage across each of the components on the breadboard add up to the voltage output from the source power supply.

2. Compare the measured source voltage to the Multisim value of this quantity. Should they be the same value ideally? Explain briefly why they are not exactly equal.

The measured source voltage and multisim source voltage vary by 6%. Multisim is a simulation where the values will always be exact, while the measured values were calculated using physical equipment which can degrade over time.

3. Does hand calculations, Multisim simulation, and breadboard measurements agree? Explain.

The calculations, simulations, and breadboard all match the overall trends of the general standard values we would expect. Some voltage was lost in the real breadboard due to real-world forces. True values are rarely 100% accurate to the simulations so our error seems correct.

4. Does percent error remains under 5%. If the percent error is greater than 5% then explain the reason for the error.

Our percent error was above 5%. This may be due to the resistance within the breadboard and the extra jumper cables we used in the experiment. Also, it could be caused by the resistance from the DMM to the breadboard through additional jumper cables. Finally, the resistor has its own resistance.

Conclusion:

We made some errors with the circuit board by not placing the LED lights correctly but we managed to learn our mistakes. We were able to identify the resistor and do the calculations correctly even when we did some miscalculations but we double checked. What we learned is on how to create a circuit reactor and connect the LED lights to get measurements for the resistors.